

After 12 MOB	After 18 MOB								
	Delinquency	No Due	0-29 DPD	30-59 DPD	60-89 DPD	90-119 DPD	120-149 DPD	150+ DPD	Total
	No Due	109994	8344	4994	3344	1694	1094	494	129958
	0-29 DPD	45494	6694	3344	1694	1694	844	494	60258
	30-59 DPD	19194	4994	3344	694	494	194	1194	30108
	60-89 DPD	5694	4994	1694	344	1694	1344	844	16608
	90-119 DPD	3344	844	494	194	694	3344	1694	10608
	120-149 DPD	494	194	194	144	144	844	1994	4008
	150+ DPD	94	64	49	39	9	4	3494	3753

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

28.69 to 29.74

Question Number : 358 **Question Id :** 6406531045780 **Question Type :** SA

Correct Marks : 1

Question Label : Short Answer Question

What is the Roll Forward rate for 60-89 DPD after 18 MOB in ?(in %)

After 12 MOB	After 18 MOB								
	Delinquency	No Due	0-29 DPD	30-59 DPD	60-89 DPD	90-119 DPD	120-149 DPD	150+ DPD	Total
	No Due	109994	8344	4994	3344	1694	1094	494	129958
	0-29 DPD	45494	6694	3344	1694	1694	844	494	60258
	30-59 DPD	19194	4994	3344	694	494	194	1194	30108
	60-89 DPD	5694	4994	1694	344	1694	1344	844	16608
	90-119 DPD	3344	844	494	194	694	3344	1694	10608
	120-149 DPD	494	194	194	144	144	844	1994	4008
	150+ DPD	94	64	49	39	9	4	3494	3753

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

22.82 to 23.92

Game Theory

Section Id : 64065375585

Section Number : 15

Section type : Online

Mandatory or Optional : Mandatory

Number of Questions :	15
Number of Questions to be attempted :	15
Section Marks :	45
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653161242
Question Shuffling Allowed :	No

Question Number : 359 Question Id : 6406531045789 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : GAME THEORY AND STRATEGY (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406533530599. ✓ YES

6406533530600. ✗ NO

Sub-Section Number :	2
Sub-Section Id :	640653161243
Question Shuffling Allowed :	No

Question Id : 6406531045790 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (360 to 366)
Question Label : Comprehension

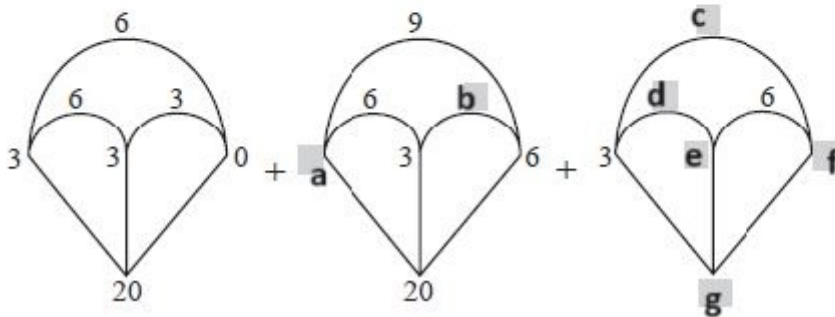
Consider the game $(N; v)$:

$$N = \{1, 2, 3\} \quad v(1) = 9 \quad v(2) = 6 \quad v(3) = 12 \quad v(\emptyset) = 0$$

$$v(1, 2) = 15 \quad v(1, 3) = 24 \quad v(2, 3) = 18$$

$$v(1, 2, 3) = 60$$

The game is split into the three games below, whose sum is the original game. Supply the missing values.



Based on the above data, answer the given subquestions

Sub questions

Question Number : 360 Question Id : 6406531045791 Question Type : SA

Correct Marks : 0.5

Question Label : Short Answer Question

Enter the Value a = _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

3

Question Number : 361 Question Id : 6406531045792 Question Type : SA

Correct Marks : 0.5

Question Label : Short Answer Question

Enter the Value b = _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

9

Question Number : 362 Question Id : 6406531045793 Question Type : SA

Correct Marks : 0.5

Question Label : Short Answer Question

Enter the Value c = _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

9

Question Number : 363 Question Id : 6406531045794 Question Type : SA

Correct Marks : 0.5

Question Label : Short Answer Question

Enter the Value d = _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

3

Question Number : 364 Question Id : 6406531045795 Question Type : SA

Correct Marks : 0.5

Question Label : Short Answer Question

Enter the Value e = _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0

Question Number : 365 Question Id : 6406531045796 Question Type : SA

Correct Marks : 0.5

Question Label : Short Answer Question

Enter the Value f = _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

6

Question Number : 366 **Question Id :** 6406531045797 **Question Type :** SA

Correct Marks : 0.5

Question Label : Short Answer Question

Enter the Value g = _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

20

Sub-Section Number :

3

Sub-Section Id :

640653161244

Question Shuffling Allowed :

No

Question Id : 6406531045798 **Question Type :** COMPREHENSION **Sub Question Shuffling**

Allowed : No **Group Comprehension Questions :** No **Question Pattern Type :** NonMatrix

Question Numbers : (367 to 369)

Question Label : Comprehension

Calculate the Shapley value of the following game:

$$N = \{1, 2, 3\} \quad v(1) = 6 \quad v(2) = 12 \quad v(3) = 18$$

$$v(1, 2) = 12 \quad v(1, 3) = 12 \quad v(2, 3) = 24$$

$$v(1, 2, 3) = 48 \quad v(\phi) = 0$$

Suppose the Shapley value is (s_1, s_2, s_3) , then

Based on the above data, answer the given subquestions

Sub questions

Question Number : 367 **Question Id :** 6406531045799 **Question Type :** SA

Correct Marks : 2

Question Label : Short Answer Question

Enter the Value s_1 = _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

11

Question Number : 368 Question Id : 6406531045800 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

Enter the Value $s_2 =$ _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

14

Question Number : 369 Question Id : 6406531045801 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

Enter the Value $s_3 =$ _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

23

Question Id : 6406531045825 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (370 to 372)

Question Label : Comprehension

Suppose a search engine has two ad slots that it can sell. Slot **a** has a clickthrough rate of 10 and slot **b** has a clickthrough rate of 5. There are three advertisers who are interested in these slots. Advertiser **x** values click at 3 per click, advertiser **y** values click at 2 per click, and advertiser **z** values click at 1 per click.

Based on the above data, answer the given subquestions

Sub questions

Question Number : 370 Question Id : 6406531045826 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

Total valuation of socially optimal allocation will be _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

40

Question Number : 371 Question Id : 6406531045827 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

VCG price for slot **a** will be _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

15

Question Number : 372 Question Id : 6406531045828 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

VCG price for slot **b** will be _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

5

Sub-Section Number : 4

Sub-Section Id : 640653161245

Question Shuffling Allowed : No

Question Id : 6406531045802 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (373 to 374)

Question Label : Comprehension

A man dies, leaving an estate worth 500 units. The deceased has two creditors; one's claim is 400 units, and the other's claim, 300 units. The division of the estate between them is carried out according to the contested garment principle.

Based on the above data, answer the given subquestions

Sub questions

Question Number : 373 Question Id : 6406531045803 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Creditor 1 with claim 400 units gets

Options :

6406533530611. ✖ 350

6406533530612. ✔ 300

6406533530613. ✖ 200

6406533530614. ✖ 250

Question Number : 374 Question Id : 6406531045804 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Creditor 2 with claim 300 units gets

Options :

6406533530615. ✖ 150

6406533530616. ✖ 300

6406533530617. ✔ 200

6406533530618. ✖ 250

Question Id : 6406531045815 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (375 to 376)

Question Label : Comprehension

Five bidders with private valuations uniformly distributed between 0 and 1 participate in a sealed bid first price auction. Assuming that everyone is playing the symmetric equilibrium bidding strategy answer the given subquestions:

Sub questions

Question Number : 375 Question Id : 6406531045816 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Optimal bid for a bidder with valuation 0.6 will be

Options :

6406533530651. ✖ 0.4

6406533530652. ✔ 0.48

6406533530653. ✖ 0.5

6406533530654. ✖ 0.55

Question Number : 376 Question Id : 6406531045817 Question Type : MCQ**Correct Marks : 1**

Question Label : Multiple Choice Question

The expected revenue of the seller in this auction scenario will be

Options :

6406533530655. ✖ 4/5

6406533530656. ✖ 3/5

6406533530657. ✔ 2/3

6406533530658. ✖ 2/5

Sub-Section Number :

5

Sub-Section Id :

640653161246

Question Shuffling Allowed :

No

Question Id : 6406531045805 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (377 to 378)

Question Label : Comprehension

A machine can be in two possible states viz. Good (G) and Bad (B). It is in the Good state 90% of the time. An item produced by the machine is Defective (D), 1% of the time if it is Good and 10% of the time if it is Bad.

Based on the above data, answer the given subquestions

Sub questions**Question Number : 377 Question Id : 6406531045806 Question Type : MCQ****Correct Marks : 1**

Question Label : Multiple Choice Question

What is the probability of the item being defective

Options :

6406533530619. ✖ 0.009

6406533530620. ✖ 0.09

6406533530621. ✔ 0.019

6406533530622. ✖ 0.19

Question Number : 378 Question Id : 6406531045807 Question Type : MCQ

Correct Marks : 1.5

Question Label : Multiple Choice Question

What is the probability that the machine is good if the item is defective

Options :

6406533530623. ✖ 0.047

6406533530624. ✔ 0.47

6406533530625. ✖ 0.0047

6406533530626. ✖ 0.19

Sub-Section Number :

6

Sub-Section Id :

640653161247

Question Shuffling Allowed :

No

Question Id : 6406531045808 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (379 to 384)

Question Label : Comprehension

Consider the market for a single network good as discussed in the lecture and write a different consumer's utility function for joining the network as $U(\theta) = \theta(a + vn^e)$ capturing the idea that it may be more plausible that a user who has a higher value for the standalone component of a technology also assigns more importance to the size of its network (i.e. consumers differ in their valuation of both the standalone and the network benefits). Here, a is the standalone benefit, $v > 0$ measures the network effect, n^e is the expected number of users joining the network and θ is uniformly distributed on the unit interval.

Based on the above data, answer the given subquestions

Sub questions

Question Number : 379 Question Id : 6406531045809 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

The value $\hat{\theta}$ corresponding to an indifferent consumer for a given price p and a given expected network size n^e will be

Options :

6406533530627. ✖ $\frac{p-a}{vn^e}$

6406533530628. ✔ $\frac{p}{(a+vn^e)}$

6406533530629. ✖ $\frac{ap}{(a+vn^e)}$

6406533530630. ✖ None

Question Number : 380 Question Id : 6406531045810 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

The maximum value of p to have buyers will be

Options :

6406533530631. ✔ $a + vn^e$

6406533530632. ✖ vn^e

6406533530633. ✖ $\frac{a}{vn^e}$

6406533530634. ✖ $\frac{a}{1+vn^e}$

Question Number : 381 Question Id : 6406531045811 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

The mass of buyers (n) as a function of $\hat{\theta}$ will be

Options :

6406533530635. ✖ $\hat{\theta}$

6406533530636. ✖ $1 + \hat{\theta}$

6406533530637. ✔ $1 - \hat{\theta}$

6406533530638. ✖ $\frac{1+\hat{\theta}}{1-\hat{\theta}}$

Question Number : 382 Question Id : 6406531045812 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

The willingness to pay ($p(n^e, n)$) for the n^{th} unit of the good when n^e units are expected to be sold will be

Options :

6406533530639. ✖ $(a + vn^e)$

6406533530640. ✖ $n(a + vn^e)$

6406533530641. ✖ $(a + vn^e)(1 + n)$

6406533530642. ✔ $(a + vn^e)(1 - n)$

Question Number : 383 Question Id : 6406531045813 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

$p(n^e, n)$ is

Options :

6406533530643. ✖ increasing function of n

6406533530644. ✔ decreasing function of n

6406533530645. ✖ decreasing function of n^e

6406533530646. ✔ increasing function of n^e

Question Number : 384 Question Id : 6406531045814 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

The fulfilled-expectations demand curve in this setting will be

Options :

6406533530647. ✖ $(a + vn)$

6406533530648. ✖ $n(a + vn)$

6406533530649. ✖

$$(a + vn)(1 + n)$$

6406533530650. ✓ $(a + vn)(1 - n)$

Sub-Section Number : 7

Sub-Section Id : 640653161248

Question Shuffling Allowed : No

Question Id : 6406531045818 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (385 to 386)

Question Label : Comprehension

Synergies: Two division managers can invest time and effort in creating a better working relationship. Each invests $e_i \geq 0$, and if both invest more then both are better off, but it is costly for each manager to invest. In particular, the payoff function for player i from effort levels (e_i, e_j) is $v_i(e_i, e_j) = (a + e_j)e_i - e_i^2$.

Based on the above data, answer the given subquestions

Sub questions

Question Number : 385 Question Id : 6406531045819 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Best response correspondence for player i , $BR_i(e_j)$, will be

Options :

6406533530659. ✗ $\frac{a+2e_j}{2}$

6406533530660. ✓ $\frac{a+e_j}{2}$

6406533530661. ✗ $a + e_j$

6406533530662. ✗ $\frac{a+2e_j}{3}$

Question Number : 386 Question Id : 6406531045820 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Nash equilibrium of this game will be

Options :

6406533530663. ✖ $e_1 = e_2 = \frac{a}{2}$

6406533530664. ✖ $e_1 = e_2 = 0$

6406533530665. ✖ $e_1 = e_2 = \frac{a}{3}$

6406533530666. ✔ $e_1 = e_2 = a$

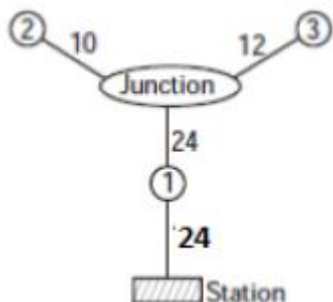
Question Id : 6406531045829 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (387 to 389)

Question Label : Comprehension

A cable TV station wants to connect three new customers to its network. The connections to the network are described in the following tree graph:



Calculate the distribution of the costs among the three players if all three decide to connect to the network.

Based on the above data, answer the given subquestions

Sub questions

Question Number : 387 Question Id : 6406531045830 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

The cost to be paid by player 1 will be _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

Question Number : 388 Question Id : 6406531045831 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

The cost to be paid by player 2 will be _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

30

Question Number : 389 Question Id : 6406531045832 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

The cost to be paid by player 3 will be _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

32

Sub-Section Number : 8

Sub-Section Id : 640653161249

Question Shuffling Allowed : No

Question Id : 6406531045821 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (390 to 392)

Question Label : Comprehension

There are 3 firms that are considering entering a new market. The payoff for each firm that enters is $\frac{150}{n}$ where n is the number of firms that enter.

The cost of entering is 62.

Based on the above data, answer the given subquestions

Sub questions

Question Number : 390 Question Id : 6406531045822 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

How many pure strategy equilibria are there in this game?

Options :

6406533530667. ✖ 1

6406533530668. ✖ 2

6406533530669. ✔ 3

6406533530670. ✖ 4

Question Number : 391 Question Id : 6406531045823 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

In one of these pure NEs, all three players enter the market

Options :

6406533530671. ✖ TRUE

6406533530672. ✔ FALSE

Question Number : 392 Question Id : 6406531045824 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

There is a symmetric mixed strategy equilibrium where all three players enter with the same probability (say p). Find the value of p

Options :

6406533530673. ✖ 1/5

6406533530674. ✖ 3/5

6406533530675. ✔ 4/5

6406533530676. ✖ 2/5

Sub-Section Number :

9

Sub-Section Id :

640653161250

Question Shuffling Allowed :

Yes

Question Number : 393 Question Id : 6406531045833 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Suppose 5 numbers are drawn independently from the uniform distribution on the interval $[0, 1]$ and then sorted from smallest to largest. The expected value of the number in second position on the list is

Options :

6406533530683. ✖ 5/6

6406533530684. ✖ 2/3

6406533530685. ✖ 1/2

6406533530686. ✔ 1/3

Sub-Section Number :

10

Sub-Section Id :

640653161251

Question Shuffling Allowed :

Yes

Question Number : 394 Question Id : 6406531045834 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

In weighted majority game [7; 5, 4, 3, 1]

Options :

6406533530687. ✖ Player 1 is null player

6406533530688. ✖ Player 2 is null player

6406533530689. ✖ Player 3 is null player

6406533530690. ✔ Player 4 is null player

Question Number : 395 Question Id : 6406531045835 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Consider following two statements:

1. Crowd can be wrong even if everyone in the crowd is rational and everyone takes the same action.
2. Aggregate behaviour of many people with limited information can sometimes produce very accurate results.

Which statement(s) is (are) true?

Options :

6406533530691. ✖ Only 1

6406533530692. ✖ Only 2

6406533530693. ✔ Both

6406533530694. ✖ None

Question Number : 396 Question Id : 6406531045836 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Choose the correct alternative

Options :

6406533530695. ✖ A stable matching system is necessarily a system under which everyone is satisfied.

6406533530696. ✖ A matching system is stable when an unmatched pair may find it beneficial to

deviate from the matching and form their own match.

6406533530697. ✖ A stable matching system is necessarily a system under which everyone is satisfied & A matching system is stable when an unmatched pair may find it beneficial to deviate from the matching and form their own match.

6406533530698. ✔ None